Analysis Tutorial 6

Vano Kakiashvili

October 13, 2025

1 Introduction

In this tutorial, we recall the definition of limits, Cauchy sequences, and solve some problems about them. We will also provide solutions to Problem 4 and 5 from the 3rd homework assignment.

2 Review

Limit of a Sequence

We call x the limit of the sequence (x_n), which is written as

$$x_n \to x \quad \text{or} \quad \lim_{n\to\infty} x_n = x,$$

if the following condition holds:

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ such that } \forall n \geq N,\ |x_n - x| < \varepsilon.$$

In other words, for every measure of closeness ε, the sequence's terms are eventually that close to the limit. The sequence (x_n) is said to converge to or tend to the limit x.

Cauchy Sequences

A sequence $x_1, x_2, x_3, \ldots$ of real numbers is called a Cauchy sequence if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ such that } \forall m, n > N,\ |x_m - x_n| < \varepsilon.$$

Similarly, one can define Cauchy sequences of rational or complex numbers.

Homework Problems Review

Problem 4 from 3rd homework: Prove that the operation of multiplication of integers is well-defined, i.e., the result does not depend on the choice of class representatives.

Problem 5 from 3rd homework:

$$\text{If } z_1, z_2 \in \mathbb{Z},\ z_1 \neq 0,\ z_2 \neq 0, \text{ then } z_1 z_2 \neq 0.$$

Problems:

Problem 1. Find the limits of the following sequences:

(a) $\dfrac{1}{n^2}$

(b) $\dfrac{n^2}{3^n}$

(c) $\sqrt{n}\left(\sqrt{n+1} - \sqrt{n}\right)$

Problem 2. Determine if the sequence

$$a_n = (-1)^{kn}, \quad k \in \mathbb{N},\, n \geq 1$$

has a limit, and find it if it exists.

Problem 3. Determine if the sequence

$$a_n = \left(-\frac{1}{2}\right)^{kn}, \quad k \in \mathbb{N},\, n \geq 1$$

has a limit, and find it if it exists.

Problem 4. Prove that the sequence

$$a_n = \frac{2n+1}{3n-1} \text{ is a Cauchy sequence.}$$

Problem 5. Prove that for every rational $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \varepsilon.$$

Problem 6. Prove that the limit of a sequence, if it exists, is unique.

Problem 7. Suppose that

$$\lim_{n\to\infty} a_n = L, \quad \lim_{n\to\infty} b_n = M,$$

and let k be a constant. Then

$$\lim_{n\to\infty} (ka_n) = k \lim_{n\to\infty} a_n = kL$$

and

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = L + M.$$

Question: If the limit of $a_n + b_n$ is K, does this imply that $\lim(a_n + b_n) = \lim a_n + \lim b_n$?

Problem 8. Determine whether the series

$$\sum_{n=2}^{\infty} \frac{1}{\sqrt{n^2-3}}$$

converges.

Problem 9. Suppose that (x_n) is a sequence satisfying

$$|x_{n+1} - x_n| \leq (1/2)^n, \quad n = 1, 2, 3, \ldots$$

Show that (x_n) satisfies the Cauchy criterion.